Statistical Modelling & Analysis (STAT2004)
Applied Class (Week 5)

We will go through Questions 3 & 4 together in the Applied Class. Try the other questions yourself during the week and discuss your results with your tutor next week.

1. Suppose that $Y \sim \mathsf{N}(\mathbf{X}\boldsymbol{\beta}, \sigma^2 I_n)$, where $\mathbf{X}$ is an $n \times p$ matrix, $n > p$ and $\sigma^2 > 0$. If $\mathbf{X}$ has full column rank, (that is, there is no $\mathbf{c} \in \mathbb{R}^p \setminus \{\mathbf{0}\}$ such that $\mathbf{Xc} = \mathbf{0}$) the MLE is unique and given by
$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T Y.$$

   (a) Show that $\mathbf{X}^T\mathbf{X}$ is invertible if and only if $\mathbf{X}$ has full column rank.

   Solution. A square matrix $\mathbf{A}$ is invertible if and only if
$$\mathbf{Ax} = \mathbf{0} \quad \implies \mathbf{x} = \mathbf{0}.$$
   Let $\mathbf{c} \in \mathbb{R}^p \setminus \{\mathbf{0}\}$ and suppose $\mathbf{Xc} = \mathbf{0}$ so $\mathbf{X}$ does not have full column rank. Then
$$\mathbf{X}^T\mathbf{Xc} = \mathbf{X}^T\mathbf{0} = \mathbf{0}.$$
   Hence, if $\mathbf{X}$ does not have full column rank, then $\mathbf{X}^T\mathbf{X}$ is not invertible.
   Now suppose that for some $\mathbf{c} \in \mathbb{R}^p \setminus \{\mathbf{0}\}$, $\mathbf{X}^T\mathbf{Xc} = \mathbf{0}$. Then
$$\mathbf{c}^T\mathbf{X}^T\mathbf{Xc} = \mathbf{c}^T\mathbf{0} = 0.$$
   Note that
$$\mathbf{c}^T\mathbf{X}^T\mathbf{Xc} = \|\mathbf{Xc}\|^2.$$
   As the norm of a vector is zero if and only if the vector is the zero vector, we see $\|\mathbf{Xc}\| = 0$ implies $\mathbf{Xc} = \mathbf{0}$. Hence, if $\mathbf{X}^T\mathbf{X}$ is not invertible, then $\mathbf{X}$ does not have full column rank.

   (b) Let $\mathbf{c} \in \mathbb{R}^p$. An unbiased estimator of $\mathbf{c}^T\beta$ is $\mathbf{c}^T\hat{\beta}$. Is there a vector $\mathbf{c} \in \mathbb{R}^p \setminus \{\mathbf{0}\}$ for which $\mathrm{Var}(\mathbf{c}^T\hat{\beta}) = 0$?

   Solution. Let $\mathbf{d} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{c}$ so $\mathbf{c} = \mathbf{X}^T\mathbf{Xd}$. Then
$$\frac{\mathrm{Var}(\mathbf{c}^T\hat{\beta})}{\sigma^2} = \mathbf{c}^T(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{c} = (\mathbf{X}^T\mathbf{Xd})^T\mathbf{d} = \mathbf{d}^T\mathbf{X}^T\mathbf{Xd} = \|\mathbf{Xd}\|^2.$$
   As $\mathbf{X}^T\mathbf{X}$ is invertible and $\mathbf{c} \neq \mathbf{0}$, we see $\mathbf{d} \neq 0$. As $\mathbf{X}$ has full column rank, $\mathbf{Xd} \neq \mathbf{0}$. It follows that $\mathbf{c} \neq \mathbf{0}$ implies $\mathrm{Var}(\mathbf{c}^T\hat{\beta}) > 0$.

2. Consider the regression-through-the-origin model
$$Y_i = \beta x_i + \epsilon_i,$$
   where the ϵ_i are IID $\mathsf{N}(0, \sigma^2)$ and $x_1, x_2, \ldots$ is a fixed (ie non-random) sequence. Assume that $x_1 \neq 0$.

   (a) Let $\hat{\beta}_n$ denote the MLE of β from the first n observations. Write down an expression for $\hat{\beta}_n$.

   Solution. Here $\mathbf{X} = (x_1, \ldots, x_n)^T$. So $\mathbf{X}^T\mathbf{X} = \sum_{j=1}^n x_j^2$ and $\mathbf{X}^T Y = \sum_{j=1}^n x_j Y_j$. The MLE of β is
$$\hat{\beta}_n = \frac{\sum_{j=1}^n x_j Y_j}{\sum_{j=1}^n x_j^2}.$$

(b) State the distribution of $\hat{\beta}_n$.

Solution. We have seen that for the general multiple linear regression model

$$\hat{\beta} \sim \mathsf{N}\left(\beta, \sigma^2(\mathbf{X}^T\mathbf{X})^{-1}\right).$$

Hence, for this model $\hat{\beta}_n \sim \mathsf{N}\left(\beta, \sigma^2(\sum_{j=1}^n x_j^2)^{-1}\right)$

(c) Give a necessary and sufficient condition on $x_1, x_2, \ldots$ for $\hat{\beta}_n$ to be a consistent estimator of β, that is for all $\delta > 0$,

$$\lim_{n\to\infty} \mathbb{P}\left(|\hat{\beta}_n - \beta| > \delta\right) = 0$$

Solution. In Week 1, we saw that if the mean-squared error (MSE) of an estimator goes to zero, then the estimator is consistent. Since $\hat{\beta}_n$ is an unbiased estimator of β, $\text{MSE}(\hat{\beta}_n) = \text{Var}(\hat{\beta}_n)$. Hence, $\hat{\beta}_n$ is a consistent estimator of β if

$$\text{Var}(\hat{\beta}_n) = \sigma^2 \left(\sum_{j=1}^n x_j^2\right)^{-1} \to 0.$$

A sufficient condition for $\hat{\beta}_n$ to be consistent is that $\sum_{j=1}^n x_j^2 \to \infty$.
Now suppose that $\sum_{j=1}^n x_j^2$ does not tend to ∞. As $\sum_{j=1}^n x_j^2$ is non-decreasing, it must then be bounded above. Let C_1 be a finite constant such that $\sum_{j=1}^n x_j^2 \le C_1$ for all n. Then

$$\begin{aligned}
\mathbb{P}\left(|\hat{\beta}_n - \beta| > \delta\right) &= \mathbb{P}\left(\left|\frac{(\hat{\beta}_n - \beta)\sqrt{\sum_{j=1}^n x_j^2}}{\sigma}\right| > \frac{\delta\sqrt{\sum_{j=1}^n x_j^2}}{\sigma}\right) \\
&= \mathbb{P}\left(|Z| > \frac{\delta\sqrt{\sum_{j=1}^n x_j^2}}{\sigma}\right) \qquad \text{where } Z \sim \mathsf{N}(0,1), \\
&\ge \mathbb{P}\left(|Z| > \frac{\delta\sqrt{C_1}}{\sigma}\right)
\end{aligned}$$

So $\hat{\beta}_n$ cannot be consistent if $\sum_{j=1}^n x_j^2$ is bounded.
The necessary and sufficient condition for $\hat{\beta}_n$ to be a consistent estimator of β is that $\sum_{j=1}^n x_j^2 \to \infty$.

(d) Interpret the condition in part (c) in terms of the sensitivity of the mean response to changes in β. Explain why increasing the number of observations does not necessarily guarantee consistency.

Solution. The variance of the estimator is

$$\text{Var}(\hat{\beta}_n) = \frac{\sigma^2}{\sum_{j=1}^n x_j^2}.$$

For a given change in β, the corresponding change in the mean of Y_j is proportional to x_j. Thus, when $|x_j|$ is close to zero, the mean response changes very little as β changes, while the error variance remains σ^2. Such an observation has little effect on the precision of the estimator. Larger values of $|x_j|$ make the mean response more sensitive to changes in β. The condition

$$\sum_{j=1}^n x_j^2 \to \infty$$

means that the accumulated squared magnitudes of the design values must grow sufficiently for the variance of $\hat{\beta}_n$ to tend to zero. Merely increasing the number of observations is not sufficient if the x_j values approach zero too quickly.

(e) Suppose $x_j \neq 0$ for all j. An alternative estimator of β is

$$\tilde{\beta}_n = n^{-1} \sum_{j=1}^{n} \frac{Y_j}{x_j}.$$

Use the Cauchy–Schwarz inequality $\left(\sum_{j=1}^n a_j b_j\right)^2 \leq \left(\sum_{j=1}^n a_j^2\right)\left(\sum_{j=1}^n b_j^2\right)$ to show $\text{MSE}(\hat{\beta}_n) \leq \text{MSE}(\tilde{\beta}_n)$. When is the MSE of the two estimators equal?

Solution. We first determine $\text{MSE}(\tilde{\beta}_n)$.

$$\mathbb{E}(\tilde{\beta}_n) = \mathbb{E}\left(n^{-1} \sum_{j=1}^{n} \frac{Y_j}{x_j}\right) = n^{-1} \sum_{j=1}^{n} x_j^{-1} \mathbb{E}(Y_j) = n^{-1} \sum_{j=1}^{n} x_j^{-1} x_j \beta = \beta.$$

$$\text{Var}(\tilde{\beta}_n) = \text{Var}\left(n^{-1} \sum_{j=1}^{n} \frac{Y_j}{x_j}\right) = n^{-2} \sum_{j=1}^{n} x_j^{-2} \text{Var}(Y_j) = \sigma^2 n^{-2} \sum_{j=1}^{n} x_j^{-2}$$

So $\tilde{\beta}_n$ is an unbiased estimator of β and

$$\text{MSE}(\tilde{\beta}_n) = \sigma^2 n^{-2} \sum_{j=1}^{n} x_j^{-2}.$$

We want to show $\text{MSE}(\hat{\beta}_n) \leq \text{MSE}(\tilde{\beta}_n)$. The MSE of $\hat{\beta}_n$ is determined in part (c). So we want to show

$$\left(\sum_{j=1}^{n} x_j^2\right)^{-1} \leq n^{-2} \sum_{j=1}^{n} x_j^{-2},$$

which is equivalent to

$$n^2 \leq \left(\sum_{j=1}^{n} x_j^2\right)\left(\sum_{j=1}^{n} x_j^{-2}\right).$$

This follows from the Cauchy-Schwarz inequality upon taking $a_j = x_j$ and $b_j = x_j^{-1}$.
Equality in the Cauchy–Schwarz inequality holds if and only if the vectors

$$(x_1, \ldots, x_n) \quad \text{and} \quad (x_1^{-1}, \ldots, x_n^{-1})$$

are linearly dependent. Thus, there is a constant c such that $x_j = c x_j^{-1}$ for every j. Equivalently, $x_j^2 = c$ for every j. Therefore, the MSEs are equal if and only if

$$|x_1| = |x_2| = \cdots = |x_n|.$$

3. Suppose that the true regression model is

$$Y = \mathbf{X}\boldsymbol{\beta} + \gamma \mathbf{z} + \boldsymbol{\epsilon}, \qquad \boldsymbol{\epsilon} \sim \mathsf{N}_n(\mathbf{0}, \sigma^2 \mathbf{I}_n),$$

where $\mathbf{X}$ is an $n \times p$ matrix with full column rank and $\mathbf{z} \in \mathbb{R}^n$ contains the observed values of an additional explanatory variable.
Suppose that $\mathbf{z}$ is omitted and the model containing only the explanatory variables in $\mathbf{X}$ is fitted. The resulting estimator is

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T Y.$$

(a) Find $\mathbb{E}(\hat{\boldsymbol{\beta}})$ under the true model.

Solution. Substituting the true model into the estimator gives

$$\begin{aligned}\hat{\boldsymbol{\beta}} &= (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T Y \\ &= (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\left(\mathbf{X}\boldsymbol{\beta} + \gamma\mathbf{z} + \boldsymbol{\epsilon}\right) \\ &= \boldsymbol{\beta} + \gamma(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{z} + (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\boldsymbol{\epsilon}.\end{aligned}$$

Since $\mathbb{E}(\boldsymbol{\epsilon}) = \mathbf{0}$,

$$\mathbb{E}(\hat{\boldsymbol{\beta}}) = \boldsymbol{\beta} + \gamma(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{z}.$$

Thus, the bias resulting from the omission of $\mathbf{z}$ is

$$\text{Bias}(\hat{\boldsymbol{\beta}}) = \gamma(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{z}.$$

(b) Suppose that $\mathbf{z}$ is orthogonal to every column of $\mathbf{X}$. Show that $\hat{\boldsymbol{\beta}}$ is an unbiased estimator of $\boldsymbol{\beta}$.

Solution. The vector $\mathbf{z}$ is orthogonal to every column of $\mathbf{X}$ if and only if

$$\mathbf{X}^T\mathbf{z} = \mathbf{0}.$$

It follows from part (a) that

$$\mathbb{E}(\hat{\boldsymbol{\beta}}) = \boldsymbol{\beta} + \gamma(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{0} = \boldsymbol{\beta}.$$

Therefore, $\hat{\boldsymbol{\beta}}$ is an unbiased estimator of $\boldsymbol{\beta}$.

(c) Suppose that the fitted model contains an intercept. What does the condition that $\mathbf{z}$ is orthogonal to the intercept column imply about the observed values in $\mathbf{z}$?

Solution. If the fitted model contains an intercept, one column of $\mathbf{X}$ is

$$\mathbf{1}_n = (1, \ldots, 1)^T.$$

Orthogonality of $\mathbf{z}$ to the intercept column requires

$$\mathbf{1}_n^T\mathbf{z} = \sum_{i=1}^{n} z_i = 0.$$

Equivalently, $\bar{z} = 0$. Thus, the observed values of the omitted explanatory variable must be centred around zero.

(d) Let

$$\mathbf{H} = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T$$

be the influence matrix for the fitted model, and let

$$\mathbf{e} = Y - \mathbf{X}\hat{\boldsymbol{\beta}}$$

be the residual vector. Assume that $\mathbf{z}$ is orthogonal to every column of $\mathbf{X}$.

i. Show that $\mathbf{Hz} = \mathbf{0}$.

ii. Find an expression for $\mathbf{e}$.

iii. Find $\mathbb{E}(\mathbf{e})$ and $\text{Var}(\mathbf{e})$.

iv. Explain what these results show about the effect of omitting $\mathbf{z}$.

Solution. Since $\mathbf{z}$ is orthogonal to every column of $\mathbf{X}$,

$$\mathbf{X}^T\mathbf{z} = \mathbf{0}.$$

Therefore,

$$\mathbf{H}\mathbf{z} = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{z} = \mathbf{0}.$$

The residual vector can be written as

$$\mathbf{e} = (\mathbf{I}_n - \mathbf{H})Y.$$

Substituting the true model gives

$$\begin{aligned}\mathbf{e} &= (\mathbf{I}_n - \mathbf{H})\left(\mathbf{X}\boldsymbol{\beta} + \gamma\mathbf{z} + \boldsymbol{\epsilon}\right)\\ &= (\mathbf{I}_n - \mathbf{H})\mathbf{X}\boldsymbol{\beta} + \gamma(\mathbf{I}_n - \mathbf{H})\mathbf{z} + (\mathbf{I}_n - \mathbf{H})\boldsymbol{\epsilon}.\end{aligned}$$

Since $\mathbf{H}\mathbf{X} = \mathbf{X}$ and $\mathbf{H}\mathbf{z} = \mathbf{0}$,

$$\mathbf{e} = \gamma\mathbf{z} + (\mathbf{I}_n - \mathbf{H})\boldsymbol{\epsilon}.$$

Consequently,

$$\mathbb{E}(\mathbf{e}) = \gamma\mathbf{z}.$$

Also,

$$\begin{aligned}\mathrm{Var}(\mathbf{e}) &= (\mathbf{I}_n - \mathbf{H})\,\mathrm{Var}(\boldsymbol{\epsilon})(\mathbf{I}_n - \mathbf{H})^T\\ &= \sigma^2(\mathbf{I}_n - \mathbf{H})(\mathbf{I}_n - \mathbf{H})^T.\end{aligned}$$

Since $\mathbf{H}$ is symmetric and idempotent, $\mathbf{I}_n - \mathbf{H}$ is also symmetric and idempotent. Hence,

$$\mathrm{Var}(\mathbf{e}) = \sigma^2(\mathbf{I}_n - \mathbf{H}).$$

Thus,

$$\mathbf{e} \sim \mathsf{N}\left(\gamma\mathbf{z}, \sigma^2(\mathbf{I}_n - \mathbf{H})\right).$$

Although omitting $\mathbf{z}$ does not bias $\hat{\boldsymbol{\beta}}$, it does affect the residuals. In particular,

$$\mathbb{E}(e_i) = \gamma z_i.$$

The omitted effect remains as a systematic component of the residuals rather than being absorbed into the estimated coefficients of the included explanatory variables. Consequently, a plot of the residuals against $\mathbf{z}$ would tend to display a linear pattern when $\gamma \neq 0$.

4. The `sat` data in the `faraway` package contain state-level educational expenditure and SAT results for the 50 US states in 1994–95. Fit the model

$$\texttt{total} = \beta_0 + \beta_1\texttt{expend} + \beta_2\texttt{salary} + \beta_3\texttt{ratio} + \beta_4\texttt{takers} + \epsilon.$$

Use appropriate diagnostic quantities and plots to answer the following questions. For every plot displayed, explain what feature is relevant and what conclusion you draw. Do not display a plot unless you have something substantive to say about it.

```
library(faraway)
?sat
sat_lm <- lm(total ~ expend + salary + ratio + takers, data = sat)
```

(a) Which state has the largest Cook's distance? Report its Cook's distance, leverage and standardised residual. Explain why this state is influential.

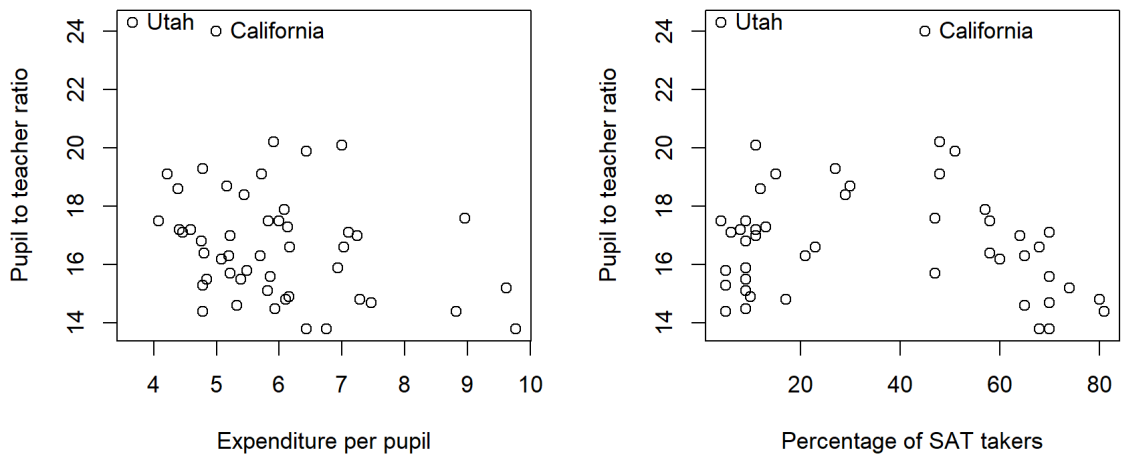
Solution. The following code obtains the diagnostic quantities.

```
sat_diagnostics <- data.frame(
  rstandard = rstandard(sat_lm),
  leverage = hatvalues(sat_lm),
  cooks.distance = cooks.distance(sat_lm),
  row.names = rownames(sat)
)
sat_diagnostics[order(sat_diagnostics$cooks.distance, decreasing = TRUE), ]
```

Utah has the largest Cook's distance. The relevant values are approximately

$$\text{Cook's distance} = 0.472, \qquad \text{leverage} = 0.292, \qquad \text{standardised residual} = 2.390.$$

Utah is influential because it combines high leverage with a large residual. Its standardised residual and leverage together produce the largest Cook's distance. Note that the values of the covariates are unusual: it has the lowest expenditure per pupil, the highest pupil-to-teacher ratio, and one of the lowest percentages of SAT takers. The observed total SAT score for Utah is also substantially larger than the value predicted by the fitted model, as indicated by the residual.



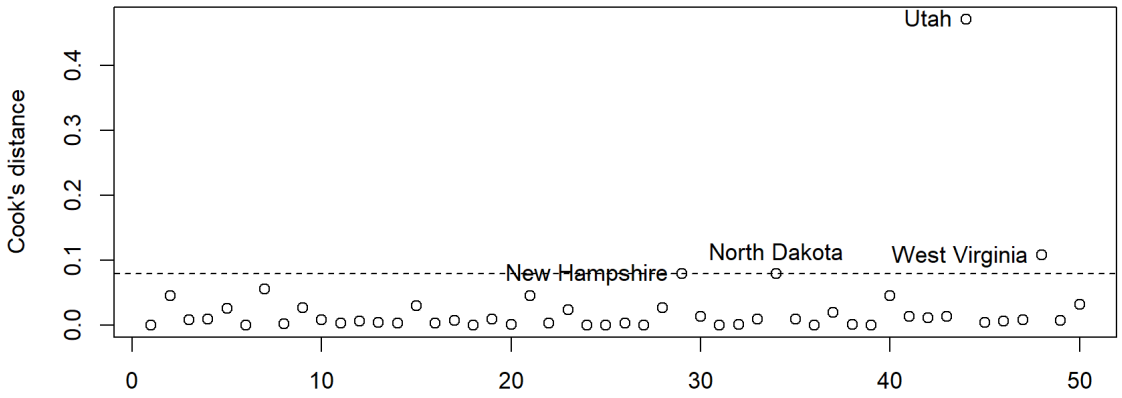

```
par(mfrow=c(1,2))
plot(sat$expend, sat$ratio,
     xlab="Expenditure per pupil",
     ylab="Pupil to teacher ratio")
identify(sat$expend, sat$ratio, labels=row.names(sat))

plot(sat$takers, sat$ratio,
     xlab="Percentage of SAT takers",
     ylab="Pupil to teacher ratio")
identify(sat$takers, sat$ratio, labels=row.names(sat))
```

A useful plot is Cook's distance against state. It shows that Utah is clearly separated from the remaining states.

```
plot(sat_diagnostics$cooks.distance,
     ylab="Cook's distance",
     xlab="")
abline(h = 4/nrow(sat), lty = 2)
identify(sat_diagnostics$cooks.distance,
         labels=rownames(sat_diagnostics))
```

The line at $4/n = 0.08$ is a guide for screening.



(b) How do the estimated coefficients change when the observation for Utah is removed from the dataset? Which coefficients change most? Do the principal conclusions from the model change?

Solution. For reference an edited version of the summary from `sat_lm` is given below.

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) 1045.9715    52.8698  19.784  < 2e-16 ***
expend         4.4626    10.5465   0.423    0.674
salary         1.6379     2.3872   0.686    0.496
ratio         -3.6242     3.2154  -1.127    0.266
takers        -2.9045     0.2313 -12.559 2.61e-16 ***
---
```

The following values give the full-model coefficient estimates minus the coefficient estimates obtained after deleting Utah.

```
dfbeta(sat_lm)["Utah",]
```

```
 (Intercept)       expend       salary        ratio       takers
-47.87443743   5.40533354  -1.45851225   4.01491196   0.02632387
```

The intercept and the coefficients of `expend`, `salary`, and `ratio` change noticeably. In particular, the estimate for `expend` changes sign and the magnitude of the estimate for `ratio` more than doubles. However, these coefficients are imprecisely estimated in the full model as indicated by the large standard errors relative to the coefficient estimates.
Deleting Utah does not materially affect the estimated coefficient of takers, so the principal conclusion concerning SAT participation remains unchanged. However, it does affect the conclusions concerning some of one other covariate. In particular, the estimated coefficient of ratio becomes more negative and the related p-value decreases to less than 0.05 after Utah is removed. Thus, conclusions concerning ratio are sensitive to the inclusion of Utah.
The full model fit after deleting Utah from the dataset is given below

```
sat_no_utah_lm <- lm(total ~ expend + salary + ratio + takers,
                     data = sat[row.names(sat) != "Utah",])
summary(sat_no_utah_lm)
---edited---
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) 1093.8460    53.4226  20.475   <2e-16 ***
expend        -0.9427    10.1922  -0.092    0.927
salary         3.0964     2.3283   1.330    0.190
ratio         -7.6391     3.4279  -2.229    0.031 *
takers        -2.9308     0.2188 -13.397   <2e-16 ***
```

(c) West Virginia has the largest absolute standardised residual. Why does deleting West Virginia have less effect on the estimated coefficients than deleting Utah?

Solution. West Virginia has

$$\text{Cook's distance} = 0.108, \qquad \text{leverage} = 0.062, \qquad \text{standardised residual} = -2.859.$$

It is the clearest response outlier: the observed total score is 932, compared with a fitted value of about 1022.5. However, its leverage is below the average leverage $p/n = 5/50 = 0.10$. The values of the covariates are therefore not unusual enough to give it a strong pull on the fitted regression surface.
Utah has a somewhat smaller absolute standardised residual, but much higher leverage. Influence depends on both residual size and leverage, so Utah has a much larger effect on the estimated coefficients.

(d) California has a relatively high leverage of 0.282. Does deleting California have a large effect on the estimated coefficients? Explain your answer using both its leverage and its standardised residual.

Solution. California has relatively high leverage, but its standardised residual is small in magnitude, around -0.572. Its Cook's distance is only about 0.026, below the guideline $4/n = 0.08$. Thus, although California is unusual in the predictor space, it lies reasonably close to the fitted regression surface so deleting it does not have a large effect on the estimated coefficients.
This illustrates that high leverage creates the potential for influence, but a high-leverage observation is not necessarily influential when its residual is small.

```
dfbeta(sat_lm)["California",]

(Intercept)      expend      salary       ratio      takers
 9.31992536  1.26239336 -0.30738250 -0.36782899 -0.00878391
```